

From Principles to Practice: Translating Human-AI Interaction Research into Effective Human-Centred AI Product Design - Findings from a Cross-industry Literature Review

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Abstract. This paper reviews and discusses how four disparate sectors, Healthcare, Automotive, Retail and Marketing, approach AI, specifically the Human-AI Interaction component of AI Products. Because each sector has unique domain challenges, the sectors also develop unique approaches. However, the review identifies a paradox: while technical capabilities are advancing, user trust is lagging due to a lack of standardized front-end design rules, which hinders adoption and commercialization. Based on the review, we developed a set of Common Practices in Human-AI Interaction and additionally propose a Cross-Industry Validation Roadmap for HAI-X in AI Products, where validation techniques are proposed from best practices of different sectors.

Keywords: Human-AI Interaction (HAI), Explainable AI (XAI), AI Product Design.

1 Introduction

The rapid integration of Artificial Intelligence (AI) into consumer products has necessitated a shift from traditional Human-Computer Interaction (HCI) to Human-AI Interaction (HAI-X). Unlike static tools, AI systems are probabilistic, adaptive, and often autonomous, creating unique challenges for user interface (UI) design and trust calibration. While recent systematic reviews have effectively mapped the academic landscape of HAI-X evaluation (Kim et al., 2023; Rong et al., 2024), there remains a critical disconnect between these theoretical guidelines and practical design rules required for market-ready AI products that are readily adapted by consumers.

Human-AI interaction explores the complex relationships and collaborations between humans and artificial intelligence systems. This field examines how people

utilize, manage, and are affected by AI technologies in various contexts. Key aspects often investigated include usability, user experience, trust, transparency, ethical considerations, and the social impact of AI. This review aims to explore key sectors where Human-AI interaction is significant, discuss common practices, and provide a foundational understanding of the topic.

2 Methodology

This study employs an Integrative Review methodology, selected for its ability to synthesize diverse data sources, including peer-reviewed experimental studies and high-impact industry reports (e.g., TELUS, 2025).

- **Search Strategy:** Databases were searched for literature published between 2019 and 2025. This window captures the emergence of Generative AI and Large Language Models (LLMs).
- **Inclusion Criteria:** Articles were selected if they (1) focused on one of the four target sectors (Healthcare, Automotive, Retail, Marketing), (2) explicitly discussed user interface (UI) design, trust calibration, or human-in-the-loop (HITL) architectures, and (3) demonstrated direct impact on end-users. Research focused solely on backend algorithmic performance was excluded. We targeted the Healthcare, Retail and Marketing sectors because of their importance to our organization, TELUS. We included the Automotive sector because of the importance of HCI in this sector, and rapid development of AI required for self-driving cars (Detjen, Henrik, et al, 2021)
- **Analysis:** Findings were synthesized to identify cross-industry patterns in transparency, explainability, and risk perception.

3 Foundational Frameworks: Ethics and Teaming

First, we review the key frameworks for the application of AI in the key sectors we reviewed.

3.1 The Transparency, Explainability and Clarity (TEC) Model and Architectural Ethics

Radanliev (2025) presents a comprehensive ethical framework for AI technologies, addressing the critical intersection of transparency, fairness, and privacy in artificial intelligence development. The study establishes systematic approaches for mitigating algorithmic biases while promoting accountability in AI systems. Through their framework, the researchers demonstrate how ethical considerations can be effectively integrated into the AI development lifecycle, providing actionable guidelines for practitioners. The findings emphasize the importance of incorporating ethical safeguards at the architectural level of AI systems, contributing to the growing body of literature on responsible AI development and governance structures.

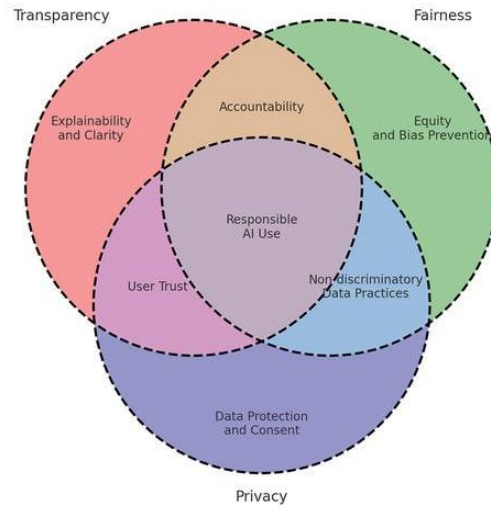


Fig. 1. Interconnected concepts of the Framework for developing AI systems that embody Transparency, Explainability and Clarity (Radanliev, 2025).

3.2 Human-Machine Teaming (HMT)

Smith (2019) presents a Human-Machine Teaming (HMT) Framework for designing ethical AI experiences in her paper "Designing Trustworthy AI: A Human-Machine Teaming Framework to Guide Development." The study emphasizes the importance of diverse teams in building trustworthy AI systems and advocates for a shared set of ethics as a foundation for development. The HMT Framework aims to bridge the gap between ethical principles and practical application in AI technology, focusing on four key themes: accountability to humans, cognizance of speculative risks and benefits, respect and security, and honesty and usability. Smith stresses the necessity of human control in AI systems, transparent decision-making processes, and the ability to override or reverse AI decisions. The research also highlights the importance of anticipating and evaluating risks to personal information and decisions affecting human life, quality of life, health, or reputation. This framework contributes to the ongoing discourse on responsible AI development by providing actionable guidelines for practitioners to align technological advancement with ethical considerations.

- Empirical survey data supports this need for control: The 2025 TELUS AI Report found that public confidence in AI decision-making nearly doubles (from 35% to 61%) when human supervision is present (TELUS, 2025).
- In the next sections, we discuss how the sectors of interest approach and apply AI within their domain.

4 Sector Analysis: Healthcare

In the healthcare sector, Human-AI interaction is transforming diagnostics, treatment planning, patient care, and administrative tasks. AI can analyze medical images, predict patient outcomes, personalize treatments, and automate routine processes. This interaction requires careful consideration of patient data privacy, AI reliability, and user training.

4.1 Medical Explainable AI (XAI)

Sheu et al. (2022) conducted a comprehensive survey on Medical Explainable AI (XAI), focusing on recent progress and challenges in the field. This study emphasizes a user-in-the-loop approach and the importance of human-machine collaboration in medical XAI systems. The authors highlight some ways in which healthcare is adopting AI while ensuring they provide the user with the highest level of transparency and input capability:

Customizable AI Explanations.

- The system offers multiple "AI robots," each providing a different style of explanation for medical diagnoses or predictions.
- Healthcare professionals can choose the explanation style that makes the most sense to them.
- This addresses the challenge of different users (e.g., doctors, nurses, specialists) having varied preferences for how they understand AI outputs.

Visual Explanation Techniques.

- Uses methods like Class Activation Mapping (CAM), Grad-CAM (Gradient-weighted Class Activation Mapping), and network dissection to create visual representations of AI decisions.
- For example, heat maps highlight the most relevant areas on medical images (like X-rays) that influenced the AI's diagnosis. This helps medical professionals quickly understand what the AI is "looking at" when making decisions.



Fig. 2. Explainable DCNN for CXR analysis and COVID-19 pneumonia classification (Sheu et al., 2022).

User Feedback Loop.

- Implements a rating system (5-point Likert scale) for users to evaluate the helpfulness of AI explanations.
- Includes a feedback box for open-ended suggestions from healthcare professionals.
- This continuous feedback helps improve the AI system over time, making it more aligned with user needs.

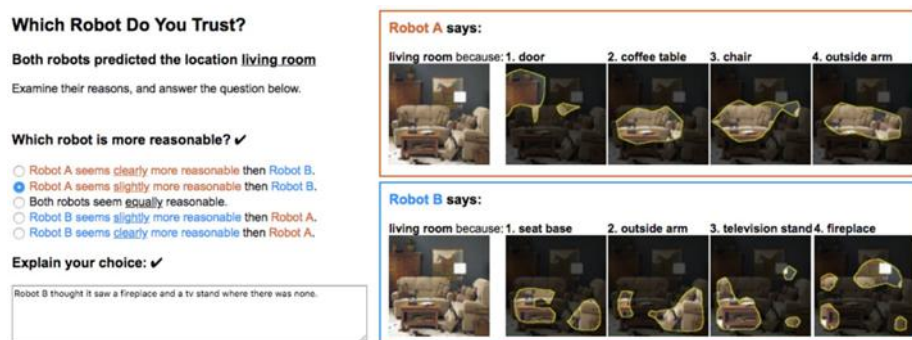


Fig. 3. Feedback system for the human rating-based evaluation (Sheu et al., 2022).

These approaches aim to make AI systems in healthcare more user-friendly and trustworthy. By allowing medical professionals to interact with AI in ways that suit their individual understanding and workflow, the system potentially improves the adoption and effective use of AI in clinical settings. This human-centered design in healthcare AI could lead to better integration of AI tools in medical decision-making processes.

4.2 Avoiding Dehumanization

Particularly noteworthy is the emergence of specialized human-AI interaction systems in mental healthcare delivery (Thieme et al., 2023). These systems demonstrate how AI can be effectively integrated into sensitive healthcare contexts while maintaining crucial human connections. For instance, in internet-delivered Cognitive Behavioral Therapy (iCBT), AI systems work alongside human supporters, typically graduate psychologists, to provide comprehensive mental health care. This hybrid approach employs conversational AI agents like Woebot for CBT delivery and Wysa for depression support, enabling more open patient self-disclosure while maintaining professional oversight.

Furthermore, in the realm of mental health and psychotherapy, the research stresses the importance of maintaining the "human-ness" in AI-assisted care delivery. The study reveals that even in current digital therapy formats, there are challenges in fostering authentic connections between care providers and patients. Care providers actively work to convey a "sense of care" through personalized communication, highlighting the irreplaceable nature of human empathy and understanding in therapeutic relationships via the following:

- **Empowering Clinicians:** Rather than replacing human providers, AI should be designed to empower and assist them in making more informed decisions.
- **Preserving Human Touch:** In mental health contexts, AI implementations should not disrupt the interpersonal relationship between patients and care providers, which is crucial for treatment success.
- **Enhancing Human Connections:** Future AI applications could focus on nurturing supporter-patient relationships, for instance, by increasing patient awareness of the supporter's role or assisting supporters in connecting more closely with patients.
- **Avoiding Dehumanization:** There's a growing concern about AI potentially "dehumanizing medicine" by reducing individuals to data points. The research emphasizes the importance of design choices that prevent oversimplification of patient experiences.
- **Balancing Efficiency and Personalization:** While AI can increase efficiency, the study cautions against standardized, machine-led communication approaches that might compromise the personal touch in healthcare interactions.

These findings suggest that the future of human-AI interaction in healthcare, especially in mental health, lies in developing systems that augment human capabilities while preserving the essential human elements of care. This approach requires a delicate balance between leveraging AI's analytical power and maintaining the irreplaceable aspects of human empathy, intuition, and personalized care.

5 Sector Analysis: Automotive

Human-AI interaction is at the forefront of autonomous driving technologies. Self-driving vehicles rely on AI to navigate, make decisions, and interact with human drivers

and pedestrians. The development of intuitive interfaces, reliable AI algorithms, and ensuring safety are critical aspects of this sector.

It is crucial to note that high-stakes interaction design is not novel to this industry. As Akamatsu et al. (2013) highlight in their review of automotive human factors, the sector has historically prioritized Man-Machine Interfaces (MMI) to manage driver workload and safety. Unlike emerging fields, automotive AI benefits from decades of established research into physical ergonomics and cognitive processing. This historical foundation means that modern HAI-X frameworks are not starting from scratch, but are rather an evolution of existing rigorous safety protocols.

5.1 The SAE Framework and Trust

The industry universally adopts the Society of Automotive Engineers (SAE) J3016B six-level automation taxonomy as the foundational framework for designing human-AI interactions (Detjen, Henrik, et al., 2021). This framework provides clear delineation of responsibilities between humans and AI systems across automation levels:

- Levels 0-2: Human remains primary operator with AI assistance
- Level 3 (Conditional Automation): Critical transition point requiring sophisticated takeover request (TOR) systems, where "the driver is a necessity, but is not required to monitor the environment at all times"
- Levels 4-5: Full AI control with optional human override capabilities, where "the vehicle is capable of performing all driving functions under certain conditions"

5.2 Trust Calibration and Transparency Systems

The industry has developed sophisticated frameworks for managing human-automation trust through transparency mechanisms (Detjen, Henrik, et al, 2021):

Three-Level Trust Model.

- **Dispositional Trust:** "Individual propensity to trust an automated system formed by culture, age, gender, and personality traits"
- **Situational Trust:** "Variance of a person's trust in a system in different situations"
- **Learned Trust:** "Result of all evaluated interactions a person had with automated systems"
- **Uncertainty Communication:** "Communicating system uncertainty can encourage the driver to calibrate his or her trust and, hence, foster the required situational awareness in safety-critical situations and increases takeover performance"
- **Reliability Communication:** Systems that communicate capabilities, as "when the vehicle's capabilities drop below an expected level, because its sensors do not work correctly the driver might overtrust the system, e.g., in a snowy environment"

5.3 Emotion-Aware and Safety-Centric Design Integration

Advanced manufacturers are implementing emotion-aware systems using cutting-edge technology to enhance both passenger comfort and vehicle safety (Chen et al., 2025). For instance, Porsche and BMW have introduced Affectiva's Emotion AI, an innovation from the MIT Media Lab that uses cameras to measure real-time emotional states of drivers and occupants. These systems aim to adjust in-car settings dynamically, such as climate control, adaptive music, ambient light, or empathic speech, to create a more comfortable journey. Similarly, BMW debuted the i Vision Dee concept at CES 2023, which includes advanced emotional interaction capabilities like personalized responses or visual expressions based on the drivers' mood.

These affective technologies operate within strict safety guidelines to manage cognitive load. The industry follows regulations issued by the National Highway Transportation Commission in 2013, which require that drivers should not spend more than 2 seconds on any operation. Furthermore, advanced safety systems incorporate real-time monitoring based on personal identification combined with physiological and psychological perception technology (Lyu & Abas, 2020). When fatigue is detected, the automobile intelligent control system alarms and wakes up the driver through speech while simultaneously controlling the car to turn on the danger warning lights and alarm the surrounding vehicles.

6 Sector Analysis: Retail

Unlike the safety-critical focus of the healthcare and automotive sectors, the retail sector leverages HAI-X to drive engagement through "Spatial Commerce" and gamified GUIs (Graphical User Interfaces). Retailers are increasingly borrowing metaphors from video game interfaces to allow users to manage complex product data.

6.1 Human-Centric Frameworks and Implementation Barriers

The foundation of successful retail AI lies in a customer-centric design approach that prioritizes adaptive learning and transparent interaction. Ebrahimi et al. (2024) identify a framework of six core categories shaping effective human-AI interaction: AI-enabled personalization, service efficiency, interaction challenges, trust/ethics, technical capabilities, and customer engagement. Central to this framework is the deployment of adaptive learning systems that evolve based on consumer behavior. The ON4OFF project terms this capability the creation of "digital twins"—detailed customer profiles that allow systems to refine offerings and communication strategies over time. By leveraging these adaptive profiles, retailers can move beyond static service models to offer dynamic, personalized support that matures with the customer relationship (Ebrahimi et al., 2024).

However, the implementation of these sophisticated frameworks faces significant hurdles regarding trust, privacy, and contextual adaptation. Ebrahimi et al. (2024) highlight that the depth of data collection required for "digital twins" raises critical ethical concerns that can inhibit customer engagement; successful adoption therefore hinges

on balancing personalization with transparent data practices. Furthermore, the effectiveness of these systems is heavily contingent on their ability to adapt to diverse cultural contexts. Research indicates that to provide truly personalized experiences, AI must be capable of navigating cultural nuances, language variations, and regional preferences, ensuring that the drive for technological efficiency does not compromise the human element of retail service.

6.2 Gamification of the GUI

Modern e-commerce platforms are increasingly adopting video game UI patterns to reduce cognitive load and enhance user "stickiness". This trend is evident in the evolution of wishlists into visual "inventory grids" reminiscent of Role-Playing Games (RPGs), where users utilize drag-and-drop functionality to organize items into slots or collections, effectively transforming static lists into active asset management tasks. Additionally, loyalty programs are incorporating "skill tree" visualizations and progress bars, such as those used by Sephora, which encourage users to "level up" to unlock tiers like free shipping or exclusive drops, creating a dopamine-driven feedback loop comparable to character progression in gaming (Tully, 2025).

6.3 Spatial Commerce and Optimization

Megha Hooli (2025) highlights that AI is redefining shopping through hyper-personalization and predictive inventory management. The integration of Deep Learning (DL) enables retailers to process vast customer datasets to offer real-time, context-aware recommendations (Hooli, 2025). Furthermore, the rise of Spatial Commerce (using WebXR or Apple Vision Pro) allows users to place 3D product models in their physical space, reducing return rates and bridging the online-offline gap.

7 Sector Analysis: Marketing

While other sectors focus on functional utility, the marketing sector reveals a critical psychological barrier to AI adoption: the "Taboo Effect" and the erosion of emotional trust. Marketing is also unique in that it is used by many sectors, and thus prevalent in consumers' lives.

7.1 The "AI Taboo" Effect

Cicek et al. (2024) provide critical empirical evidence that explicitly labeling a product or service as "AI-generated" can significantly reduce purchase intention. The "AI" label erodes emotional trust. Consumers perceive AI-generated content as lacking authenticity, "soul," and emotional depth. This is particularly damaging for hedonic products (e.g., luxury goods, hospitality, arts) where the "human touch" is a core value proposition. The negative impact is stronger for high-risk products (e.g., medical devices,

financial services) where the cost of failure is high. In these contexts, the "AI" label signals a lack of human accountability.

7.2 The "Uncanny Valley" of Advertising

Recent industry controversies highlight the risks of generative AI in creative marketing.

- Under Armour (2024): The brand faced backlash for an AI-powered commercial featuring boxer Anthony Joshua. Critics argued the ad repurposed existing footage without credit and lacked "soul," leading to accusations of "recycling" rather than creating (Das et al, 2024).
- Toys "R" Us (2024): The release of the first brand film created with OpenAI's Sora was met with negative sentiment, with audiences describing the AI-generated child protagonist as "creepy" and the overall ad as "soulless" (Thalen, 2024).
- Coca-Cola (2024/2025): Despite technical sophistication in their "Masterpiece" campaign, their subsequent AI-generated holiday ads were criticized for looking "dystopian" and lacking the warmth associated with the brand's heritage (Disotto, 2024).

7.3 The Counter-Movement: "Real Beauty"

In response to the proliferation of synthetic content, brands are leveraging Human-Centric AI positioning as a differentiator. In 2024/2025, Dove renewed its "Real Beauty" pledge, explicitly promising to never use AI to represent women in its advertising (Dove, 2024). This strategic move capitalizes on the "AI Taboo" by positioning human authenticity as a premium, ethical choice. The literature suggests a bifurcation in design strategy, use AI for optimization (backend) but prioritize humanity for connection (frontend).

8 Discussion

The review identifies a paradox: while technical capabilities are advancing, user trust is lagging due to a lack of standardized front-end design rules. To fill this "framework void" and close the commercialization gap, this study synthesizes the findings into a set of Common Practices in Human-AI Interaction. These practices should serve as the foundation for a standardized HAI-X design methodology.

8.1 User-Centered Design: Prioritizing Human Needs

User-centered design (UCD) must be the cornerstone of effective HAI-X, shifting the focus from technological capability to human utility. The analysis of the Retail and Automotive sectors confirms that AI systems are most effective when they integrate seamlessly into existing workflows rather than disrupting them. To achieve this, development must begin with comprehensive user research, employing surveys and

observational studies to map user personas and journeys, followed by an iterative process of prototyping and co-design. Furthermore, design must be contextual; as seen in Automotive UI, interfaces must adapt to environmental constraints, such as the high cognitive load experienced while driving. Finally, inclusive design principles are essential to ensure accessibility for diverse populations, preventing the technological exclusion of users with different levels of digital literacy or physical abilities.

8.2 Explainable AI (XAI): Enhancing Transparency

As evidenced by the Healthcare sector (Sheu et al., 2022), explainability is not optional; it is a prerequisite for trust. To foster effective human-AI collaboration, developers must move beyond "black box" algorithms by prioritizing interpretable models, such as decision trees, or utilizing post-hoc explanation techniques like LIME and SHAP for complex neural networks. These technical explanations must be translated into user-friendly formats, such as data visualizations or natural language descriptions that demystify the decision-making logic. Additionally, incorporating confidence metrics, such as an "80% match" indicator, helps users gauge when to rely on the system and when to exercise skepticism. This practice is critical for safety in Automotive contexts and for decision-making in Retail recommendations.

8.3 Ethical Considerations: Responsible Development

Ethical frameworks (Radanliev, 2025; Smith, 2019) must be integrated at the architectural level rather than treated as an afterthought. This requires rigorous bias mitigation strategies, including regular audits of training data and the application of fairness metrics to ensure equitable treatment across demographic groups. Beyond bias, privacy-preserving techniques like federated learning and differential privacy are essential for protecting user data. Accountability is equally critical; establishing clear lines of responsibility is vital, as the Marketing analysis demonstrates that consumers punish brands that promote the use of AI to evade accountability, known as the "AI Taboo." To oversee these complex issues, organizations should implement ethical review boards and clear policies for handling edge cases throughout the development lifecycle.

8.4 Usability Testing: Evaluating Interaction

Standardized metrics for evaluating HAI are currently lacking, necessitating a multi-faceted approach to testing. Future frameworks must move beyond simple functionality checks to include task-based testing, where users are observed completing specific goals to measure friction and error rates. These quantitative measures should be complemented by qualitative insights derived from think-aloud protocols and eye-tracking studies, which reveal how users visually process AI-generated information. Furthermore, testing must specifically address the psychological dimension of interaction. Using surveys to quantify changes in "Emotional Trust" and "Perceived Risk" after interaction is essential for understanding long-term adoption, addressing the findings of Cicek et al. (2024).

8.5 Feedback Mechanisms: Continuous Improvement

A robust feedback loop is essential for creating adaptive systems that evolve with user needs. Instead of static deployment, systems should incorporate in-app feedback tools, such as "Thumbs Up/Down" mechanisms in Retail or Customer Service, to capture real-time user sentiment. This explicit feedback, combined with implicit performance monitoring and sentiment analysis, provides the data necessary to refine model accuracy. Crucially, organizations must establish clear processes where this input directly influences model retraining. By creating a visible roadmap of how user feedback is incorporated, developers can transform the AI from a static tool into a responsive, learning partner.

8.6 Training and Education: Empowering Users

The "Commercialization Gap" is partly a literacy gap; therefore, empowering users through education is as critical as the technology itself. Effective HAI-X requires intuitive onboarding experiences, such as gamified tutorials or interactive walkthroughs that introduce users to AI capabilities in a safe environment. Beyond initial training, "just-in-time" learning resources, such as contextual tooltips, support users as they navigate complex features. As indicated by the TELUS report, increasing public AI literacy, specifically regarding the system's limitations, is crucial. When users understand not just how to use the AI, but how it functions and where it might fail, they can form realistic expectations and build sustainable trust (TELUS, 2025).

8.7 Practical Validation Roadmap

Synthesizing the distinct approaches from healthcare, automotive, retail, and marketing reveals that no single industry possesses a complete approach for HAI-X design; rather, they offer complementary strategies that address different stages of the product lifecycle. Based on these in-sector strategies and the Common Practices in Human-AI Interaction, we propose the following validation roadmap for HAI-X design in AI Products:

Table 1. The Proposed Cross-Industry Validation Roadmap for HAI-X in AI Products.

Product Phase	Critical Challenge	Validation Action (The "Rule")	Sector Origin
1. Architecture	Algorithmic Bias	Integrate "Architectural Ethics": Embed fairness metrics and privacy safeguards (e.g., Differential Privacy) before UI development begins.	General Frameworks (TEC Model)
2. Interface Design	"Black Box" Mistrust	Customizable Explainability: Offer varied explanation depths (e.g., visual heatmaps vs. text) based on user expertise.	Healthcare (Medical XAI)
3. Interaction	Cognitive Overload	Uncertainty Communication: Enforce strict latency limits (<2s) and visualize confidence intervals (e.g.,	Automotive (SAE Standards)

		"80% match") to calibrate trust.	
4. Engagement	User Drop-off	Gamified Progression: Use "skill trees" or inventory grids to reduce friction and reward user inputs, rather than static lists.	Retail (Spatial Commerce)
5. Go-to-Market	The "AI Taboo"	Implicit Framing: Avoid explicit "AI" labeling for hedonic/luxury goods to preserve emotional trust; focus on "Human-Centric" benefits.	Marketing (Psychological Barriers)
6. Deployment	Model Drift	Explicit Feedback Loops: Implement low-friction rating systems (e.g., Thumbs Up/Down) that directly trigger model retraining.	Healthcare / Retail

9 Conclusion

The rapid evolution of Artificial Intelligence has outpaced the development of standardized design rules, creating a "Commercialization Gap" between theoretical potential and market reality. This review highlights that while technical capability is necessary, it is insufficient for product success. As evidenced by the "AI Taboo" in marketing and the trust barriers in healthcare, the primary obstacles to adoption are often psychological and communicative rather than algorithmic.

To bridge this gap, this paper proposes that future HAI-X frameworks must move beyond a monolithic approach and instead adopt a hybrid strategy derived from sector-specific successes. A market-ready AI product must possess the safety protocols of the automotive industry, the empathetic explainability of healthcare diagnostics, the engagement mechanics of modern retail, and the authenticity-first framing of the marketing sector.

Ultimately, the transition from "Human-in-the-Loop" to "Human-Centred Product Design" requires a fundamental shift: viewing AI not as an autonomous agent to be monitored, but as a collaborative partner designed to empower. By implementing the Validation Roadmap outlined in this study, practitioners can navigate the complexities of trust, ethics, and usability, ensuring that AI technologies serve to augment, rather than diminish, the human experience.

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References

1. Akamatsu, M., Green, P., Bengler, K.: Automotive technology and human factors research: Past, present, and future. *International Journal of Vehicular Technology* 2013, 526180 (2013)
2. Chen, X., Wang, X., Fang, C., Fang, L., Gong, W., Liu, C., Wang, S.J.: Emotion-aware Design in Automobiles: Embracing Technology Advancements to Enhance Human-vehicle Interaction. In: *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*, pp. 1–18. ACM, New York (2025)
3. Cicek, M., Gursay, D., Lu, L.: Adverse impacts of revealing the presence of “Artificial Intelligence (AI)” technology in product and service descriptions on purchase intentions: the mediating role of emotional trust and the moderating role of perceived risk. *Journal of Hospitality Marketing & Management* 34(1), 1–23 (2024)
4. Detjen, H., Faltaous, S., Pfleging, B., Geisler, S., Schneegass, S.: How to increase automated vehicles’ acceptance through in-vehicle interaction design: A review. *International Journal of Human-Computer Interaction* 37(4), 308–330 (2021)
5. Disotto, J.-A.: Coca-Cola’s iconic “holidays are coming” ad is now a soulless and creepy dystopian nightmare made by ai. *TechRadar* (2024), <https://www.techradar.com/computing/artificial-intelligence/coca-colas-iconic-holidays-are-coming-ad-is-now-a-soulless-and-creepy-dystopian-nightmare-made-by-ai>, last accessed 2026/01/27
6. Dove: Dove marks 20 years of Real Beauty with a renewed commitment to 'Real' and pledge to never use AI to represent real women in its advertising. *PR Newswire* (2024), <https://www.prnewswire.com/news-releases/dove-marks-20-years-of-real-beauty-with-a-renewed-commitment-to-real-and-pledge-to-never-use-ai-to-represent-real-women-in-its-advertising-302111130.html>, last accessed 2026/01/27
7. Ebrahimi, A., Ebrahimi, M., Ebrahimi, N.: Analyzing the relationship between artificial intelligence and Customer Experience Improvement in the online retail industry. *Digital Transformation and Administration Innovation* 2(4), 73–81 (2024)
8. Hooli, M.: Optimizing Retail with AI: How Intelligent Systems Are Redefining Shopping Experiences. *Journal of Computer Science and Technology Studies* 7(6), 465–471 (2025)
9. Jianan, L., Abas, A.: Development of human-computer interactive interface for intelligent automotive. *International Journal of Artificial Intelligence* 7(2), 13–21 (2020)
10. Kim, S.S., Watkins, E.A., Russakovsky, O., Fong, R., Monroy-Hernández, A.: Humans, ai, and context: Understanding end-users’ trust in a real-world computer vision application. In: *Proceedings of the 2023 ACM Conference on Fairness, Accountability, and Transparency*, pp. 77–88. ACM, Chicago (2023)
11. Marcus Venu Das, Vimalan, D., Shirbeeni, A.Z., Eaton, M.: Under Armour’s AI-powered commercial accused of plagiarism. *Marketing Interactive* (2024), <https://www.marketing-interactive.com/under-armour-s-ai-powered-commercial-accused-of-plagiarism>, last accessed 2026/01/27
12. Radanliev, P.: AI ethics: Integrating transparency, fairness, and privacy in AI development. *Applied Artificial Intelligence* 39(1), 2463722 (2025)
13. Rong, Y., Leemann, T., Nguyen, T.-T., Fiedler, L., Qian, P., Unhelkar, V., Seidel, T., Kasneci, G., Kasneci, E.: Towards human-centered explainable AI: A survey of user studies for model explanations. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 46(4), 2104–2122 (2024)
14. Sheu, R.K., Pardeshi, M.S.: A survey on medical explainable AI (XAI): recent progress, explainability approach, human interaction and scoring system. *Sensors* 22(20), 8068 (2022)

15. Smith, C.J.: Designing trustworthy ai: A human-machine teaming framework to guide development. arXiv preprint arXiv:1910.03515 (2019)
16. TELUS Responsible AI, <https://www.telus.com/en/about/trust/responsible-ai>, last accessed 2026/01/27
17. Thalen, M.: Toys “R” us animated its iconic giraffe using AI-and people are upset. The Daily Dot (2024), <https://dailymdot.com/toys-r-us-ai-commercial-backlash>, last accessed 2026/01/27
18. Thieme, A., Hanratty, M., Lyons, M., Palacios, J., Marques, R.F., Morrison, C., Doherty, G.: Designing human-centered AI for mental health: Developing clinically relevant applications for online CBT treatment. *ACM Transactions on Computer-Human Interaction* 30(2), 1–50 (2023)
19. Tully, P.: Top 10 examples: Ecommerce gamification. Pimberly (2025), <https://pimberly.com/blog/top-10-ecommerce-gamification-examples-will-revolutionize-shopping/>, last accessed 2026/01/27